A coupled panel method and hydraulic network for porous airfoil aerdynamics⋆,⋆⋆

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ABSTRACT

Passive, pressure-driven flow between connected surface pores can modify aerodynamic forces without external actuation. This study develops a reduced-order aerodynamic–hydraulic framework for a NACA 0018 airfoil by coupling an inviscid source–vortex panel method with rectangular-channel flow models. Three configurations are investigated: horizontal channels connecting chordwise locations on the same surface, vertical channels connecting the lower and upper surfaces, and a combined arrangement of hydraulically independent channels. Aerodynamic coefficients are obtained by integrating the predicted surface pressure. At positive angles of attack, horizontal channels increase lift through pressure redistribution, with larger channel heights producing greater increases in lift. Vertical channels reduce the pressure difference between the two surfaces and decrease lift relative to the impermeable baseline. In the combined configuration, the channels interact through the external pressure field, yielding a response closer to that of the solid airfoil. Pressure and velocity difference fields show that the strongest flow modifications occur near the pore openings. These results identify channel orientation and pore connections as key factors governing the aerodynamic response of porous airfoils. The framework supports preliminary design screening and provides inviscid predictions for subsequent assessment using viscous simulations and experiments.

1. Introduction

Improving airfoil performance without relying on powered control systems remains an important objective in aerospace and energy applications. Conventional high-lift and flow-control devices can be effective, but they may introduce mechanical complexity, additional mass, maintenance requirements, and changes that disrupt the smooth external surface of the airfoil. Passive flow-control concepts provide an alternative strategy in which natural pressure differences around an aerodynamic body drive flow redistribution. Porous surfaces, permeable regions, localised pores, and internal channels employ this principle by permitting fluid transport between regions of unequal pressure. Their effectiveness, however, depends on pore placement, the connections between pores, and the strength of the interaction between the resulting surface-normal flow and the external aerodynamic field.

Porosity does not necessarily improve aerodynamic performance. Porosity extending along the entire airfoil chord, as well as other distributed porous arrangements, can reduce the rate of lift increase with angle of attack and produce competing effects on drag [1, 2]. Experiments and numerical studies further show that performance depends on the location of the porous region and the permitted internal flow directions [3, 4, 5]. These findings motivate the use of selectively connected pores to control pressure redistribution.

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For an airfoil containing individual channels, the aerodynamic effect depends on which pressure regions are connected. A channel connecting pores on the same airfoil surface transfers fluid between different chordwise locations, whereas a channel connecting the pressure and suction surfaces permits flow between the two sides of the airfoil. These arrangements can produce different changes in surface pressure even when their pore areas are similar. Consequently, porosity alone does not specify the design: channel connectivity and resistance must also be represented. This distinction is particularly relevant when several channels operate simultaneously, because their influence on the external pressure field can alter the flow through neighbouring channels.

Thin-airfoil models describe steady and unsteady porous-airfoil behaviour by relating pressure differences to flow through distributed permeability [6, 7], although accuracy can decrease for small porous regions or low flow resistance [8]. These formulations provide efficient aerodynamic predictions but do not explicitly resolve airfoil thickness or finite-length channels connecting selected surface pores. Their local porous boundary conditions are useful for distributed permeable surfaces, whereas modelling individual channels requires the pressure at each connected pore and a relation between pressure difference, channel geometry, and flow rate.

Passive internal connections have already been investigated for aerodynamic control. Acarer reported improved peak lift-to-drag ratio using passive slots [9], while Kestel et al. modified separated vortex flow through passive internal channels [10]. Bahadi et al. examined curved slots for improved performance at high angles of attack [11]. These studies establish the potential of pressure-driven internal

flow. The present work builds on this principle by developing a general reduced-order framework for multiple pairs of pores connected by independent channels, enabling different arrangements to be evaluated consistently.

Porous-region optimisation has also been investigated for static-stall control [12]. For individually connected pores, the design variables additionally include pore locations, channel dimensions, and the choice of same-surface or cross-surface connections. Changes in channel length and height modify hydraulic resistance, while changes in pore position modify the available pressure difference. Both effects must therefore be considered together. Exploring these combinations solely through viscous simulations or experiments is costly, motivating a computationally efficient model for preliminary assessment before detailed analysis of selected configurations.

Potential-flow panel methods offer an efficient means of calculating surface pressures on airfoils of finite thickness without discretising the surrounding fluid domain [13, 14, 15]. In a two-dimensional source–vortex formulation, the source strengths enforce the surface-normal boundary condition, while circulation and the trailing-edge Kutta condition determine the lifting solution. This structure accommodates porous flow by replacing the impermeability condition locally with a prescribed pore-normal velocity. The resulting surface velocities determine the pressure differences driving the internal channel flow. A panel formulation therefore provides a direct connection between airfoil geometry, pore flow, and aerodynamic forces at a cost suitable for repeated design evaluations.

Although the panel equations are linear for prescribed pore velocities, the complete aero–hydraulic problem is generally nonlinear. Surface pressure depends quadratically on velocity, and changing the channel flow changes both the pore velocities and the pressures driving that flow. This feedback requires a consistent coupled solution even when a linear laminar hydraulic-resistance relation is used. Treating channel flow as a fixed input would omit this interaction. The present framework instead updates the internal flow and external pressures until the coupled equations satisfy the prescribed convergence criterion.

Established viscous–inviscid methods combine an outer-flow solution with boundary-layer equations to represent additional aerodynamic effects [16]. XFOIL uses this approach with a panel-based inviscid solution and transition modelling [17]. Here, the external solver remains inviscid to isolate the aero–hydraulic interaction at low computational cost, and XFOIL provides a reference for the solid-airfoil baseline. This comparison assesses the external aerodynamic calculation; validation of the coupled porous model requires separate evidence for the internal flow and its interaction with the surrounding fluid.

The present study addresses the need for a reduced-order framework that explicitly represents finite-length channels between selected pairs of pores and couples their pressure-driven flow to the external aerodynamic solution. A two-dimensional source–vortex panel method with a Kutta condition is coupled iteratively to a hydraulic-resistance model for rectangular channels. Three NACA 0018 configurations are examined: four chordwise same-surface channels, four pressure-to-suction-surface channels, and a combined arrangement containing eight hydraulically independent channels. Channel heights of 8, 9, and 10 mm are assessed over angles of attack from $-5°$ to $15°$ at a chord Reynolds number of 5×10^5.

The combined arrangement permits the aerodynamic interaction between channel types to be assessed without introducing internal junctions. Although each channel is hydraulically independent, all channels influence a shared external pressure field. Comparing the three configurations therefore examines how their effects combine and whether connectivity changes the sensitivity to channel height and angle of attack.

The contribution is a consistent framework for comparing how channel connectivity affects internal flow, surface pressure, lift, pitching moment, and pressure-based streamwise force. The method supports preliminary design screening. Since it does not resolve boundary layers, skin friction, separation, or stall, its predictions are interpreted as inviscid trends requiring subsequent viscous simulation and experimental validation.

2. Coupled aero-hydraulic methodology

2.1. Model overview

A two-dimensional aero-hydraulic framework is developed to predict the interaction between the external flow around an airfoil and the internal flow through channels connecting selected surface openings. The external flow is assumed steady, incompressible, inviscid and irrotational, apart from the circulation represented by the panel formulation. Internal channel flow is described using a one-dimensional pressure–flow relation for parallel-wall channels.

Each channel carries a flow rate per unit span q'_k, defined as positive from opening 1 to opening 2. Channels are hydraulically independent: intersections in their two-dimensional representation do not form connected junctions. However, their flows interact through the external pressure field. External boundary layers, separation, channel entrance-region development and outlet-jet mixing are not resolved.

The solution procedure is summarised in Fig. 1. The linear aerodynamic equations are reduced exactly before solving the nonlinear channel-flow equations. Newton linearisation is applied at each iteration, and the final candidate is checked using the original aerodynamic and hydraulic models.

2.2. External source–vortex panel formulation

The airfoil boundary is represented by N straight panels, ordered clockwise. The midpoint of each panel is used as

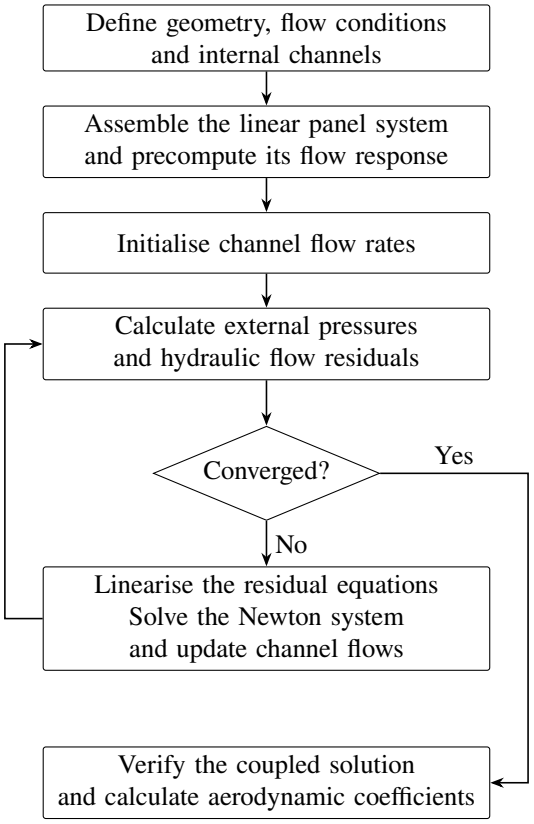


Figure 1: Coupled aero-hydraulic solution procedure. The linear panel equations are reduced exactly, while the nonlinear channel-flow residuals are linearised at each Newton iteration. Candidate solutions are independently verified before acceptance.

a collocation point. Each panel carries a constant source strength λ_i, while a single vortex strength γ is shared by all panels.

The source strengths determine the normal-velocity distribution, while the vortex strength permits circulation to develop. The unknown aerodynamic vector is

$$\mathbf{x} = \begin{bmatrix} \lambda_1 & \cdots & \lambda_N & \gamma \end{bmatrix}^T. \tag{1}$$

Let $\mathbf{t}_i$ and $\mathbf{n}_i$ denote the unit tangent and outward unit normal of panel i, respectively. The normal boundary condition is

$$\mathbf{V}_i \cdot \mathbf{n}_i = V_{n,i}. \tag{2}$$

The prescribed value $V_{n,i}$ is zero on impermeable panels. At a channel opening, it represents the panel-average suction or blowing velocity generated by the internal flow.

Using the influence-coefficient convention implemented in the solver, the normal-velocity equation is

$$\begin{aligned} &\frac{\lambda_i}{2} + \frac{1}{2\pi}\sum_{\substack{j=1\\ j\neq i}}^{N} I_{ij}\lambda_j - \frac{\gamma}{2\pi}\sum_{j=1}^{N} K_{ij} \\ &\quad = V_{n,i} - V_\infty \cos\beta_i. \end{aligned} \tag{3}$$

Here, β_i is the angle between the freestream direction and the outward panel normal. The coefficients I_{ij} and K_{ij} describe the normal influence of source and vortex distributions on panel j at collocation point i. The source self-influence appears explicitly as $\lambda_i/2$.

The corresponding tangential velocity is

$$\begin{aligned} V_{t,i} = V_\infty \sin\beta_i &+ \frac{1}{2\pi}\sum_{j=1}^{N} J_{ij}\lambda_j \\ &+ \frac{\gamma}{2} - \frac{\gamma}{2\pi}\sum_{j=1}^{N} L_{ij}. \end{aligned} \tag{4}$$

The coefficients J_{ij} and L_{ij} are the corresponding tangential influence coefficients. All influence coefficients are evaluated by integrating over the straight panels.

The N normal-velocity equations are supplemented by a trailing-edge Kutta condition. With the adopted clockwise boundary ordering, the tangential velocities on the two trailing-edge panels satisfy

$$V_{t,1} + V_{t,N} = 0. \tag{5}$$

This provides the additional equation required to determine the shared vortex strength.

For M channels, define the flow vector as

$$\mathbf{q} = \begin{bmatrix} q'_1 & \cdots & q'_M \end{bmatrix}^T. \tag{6}$$

The prescribed normal velocities are expressed as

$$\mathbf{V}_n = \mathbf{M}\mathbf{q}, \tag{7}$$

where $\mathbf{M}$ is the conservative transfer operator defined in Section 2.3.

The complete aerodynamic system is

$$\mathbf{A}\mathbf{x} = \mathbf{b}_\infty + \mathbf{B}\mathbf{q}. \tag{8}$$

The matrix $\mathbf{A}$ contains the source–vortex influence coefficients and the Kutta equation. The freestream contribution is contained in $\mathbf{b}_\infty$, while

$$\mathbf{B} = \begin{bmatrix} 2\pi\mathbf{M} \\ \mathbf{0}_{1\times M} \end{bmatrix}. \tag{9}$$

The final zero row indicates that transpiration does not directly modify the right-hand side of the Kutta equation.

For a prescribed channel-flow vector, the system is solved using an LU factorisation of $\mathbf{A}$. The resulting source and vortex strengths are substituted into Eq. (4) to recover the external tangential velocity.

The pressure coefficient includes both velocity components:

$$C_{p,i} = 1 - \frac{V_{t,i}^2 + V_{n,i}^2}{V_\infty^2}. \tag{10}$$

The panel pressure is

$$p_i = p_\infty + \frac{1}{2}\rho V_\infty^2 C_{p,i}. \tag{11}$$

2.3. Conservative transfer at channel openings

Each opening is obtained by intersecting the channel strip with the discretised airfoil boundary. Its surface length generally differs from the channel height H_k because of the local surface orientation.

Let $\ell_{ik}^{(r)}$ be the overlap length between panel i and opening r of channel k. The opening length and its contribution to the panel-average normal velocity are

$$\ell_k^{(r)} = \sum_i \ell_{ik}^{(r)}, \tag{12}$$

$$V_{n,i}^{(k,r)} = s_r q_k' \frac{\ell_{ik}^{(r)}}{\ell_k^{(r)} S_i}, \tag{13}$$

where S_i is the panel length, $s_1 = -1$ and $s_2 = +1$. Positive q_k' therefore produces suction at opening 1 and blowing at opening 2. This transfer conserves flow:

$$\sum_i V_{n,i}^{(k,1)} S_i = -q_k', \tag{14}$$

$$\sum_i V_{n,i}^{(k,2)} S_i = +q_k'. \tag{15}$$

Contributions from all channels are superposed. For partially covered panels, the prescribed panel-average velocity differs from the velocity over the open portion.

The opening pressure is evaluated as

$$\overline{p}_{k,r} = \frac{1}{\ell_k^{(r)}} \int_{\mathcal{O}_k^{(r)}} p(s)\,\mathrm{d}s. \tag{16}$$

The pressure reconstruction is piecewise linear through panel-centre values, with constant continuation over endpoint half-panels. Its integration defines a linear averaging operator on the panel pressures.

2.4. Internal hydraulic model

Each straight channel has height H_k and centreline length L_k, measured between its surface endpoints. Sidewall effects are neglected, and the internal flow is represented between parallel walls using a two-dimensional, per-unit-span formulation. The signed flow rate per unit span is denoted by q_k'.

The mean channel velocity, hydraulic diameter and channel Reynolds number are

$$U_k = \frac{q_k'}{H_k}, \tag{17}$$

$$D_{h,k} = 2H_k, \tag{18}$$

$$Re_{h,k} = \frac{\rho |U_k| D_{h,k}}{\mu}. \tag{19}$$

A two-regime hydraulic model is adopted, with a prescribed switching Reynolds number of 2300. The Hagen–Poiseuille relation for parallel-wall flow is used when $Re_{h,k} \le 2300$, while the Darcy–Weisbach relation with the Blasius friction factor is used when $Re_{h,k} > 2300$. This abrupt switch is a modelling assumption; no separate transition correlation or blending function is employed.

For steady, fully developed laminar flow, the axial momentum equation reduces to

$$\mu \frac{\mathrm{d}^2 u_k}{\mathrm{d}y^2} = -\frac{\Delta p_{f,k}}{L_k}, \qquad 0 \le y \le H_k, \tag{20}$$

where $\Delta p_{f,k} \ge 0$ is the magnitude of the distributed frictional pressure drop and u_k is measured in the direction of flow. Applying no-slip conditions at both walls gives the parabolic velocity profile

$$u_k(y) = \frac{\Delta p_{f,k}}{2\mu L_k} y(H_k - y). \tag{21}$$

Integration across the channel height gives the Hagen–Poiseuille relation per unit span:

$$|q_k'| = \int_0^{H_k} u_k(y)\,\mathrm{d}y = \frac{H_k^3}{12\mu L_k} \Delta p_{f,k}. \tag{22}$$

Consequently, the laminar pressure–flow relation is

$$\Delta p_{f,k} = \frac{12\mu L_k}{H_k^3} |q_k'|, \tag{23}$$

with hydraulic resistance per unit span

$$R_k' = \frac{12\mu L_k}{H_k^3}. \tag{24}$$

Thus, for fixed channel geometry and fluid properties, the laminar frictional pressure drop varies linearly with the flow-rate magnitude.

The distributed frictional pressure drop can also be expressed in Darcy–Weisbach form:

$$\Delta p_{f,k} = f_{D,k} \frac{L_k}{D_{h,k}} \frac{\rho U_k^2}{2}. \tag{25}$$

For parallel-wall laminar flow, the Darcy friction factor $f_{D,k} = 96/Re_{h,k}$ makes Eq. (25) equivalent to the Hagen–Poiseuille relation in Eq. (23).

For nonzero flow, the friction factor is therefore represented as

$$f_{D,k} = \begin{cases} \dfrac{96}{Re_{h,k}}, & 0 < Re_{h,k} \le 2300, \\ 0.3164 Re_{h,k}^{-1/4}, & Re_{h,k} > 2300. \end{cases} \tag{26}$$

The turbulent expression is the Blasius correlation for smooth-wall friction. Substitution into Eq. (25) gives

$$\Delta p_{f,k} = 0.3164 Re_{h,k}^{-1/4} \frac{L_k}{2H_k} \frac{\rho}{2} \left(\frac{q_k'}{H_k} \right)^2. \tag{27}$$

At zero flow, the frictional pressure drop is evaluated directly as zero without evaluating the singular friction-factor expression.

During hydraulic inversion, the Reynolds number is calculated from the candidate channel flow. A candidate is accepted only when it satisfies both the channel equation and the Reynolds-number condition associated with its selected regime.

2.5. Coupled formulation and nonlinear solution

For M channels, let $\mathbf{q} = (q'_1, \dots, q'_M)^\mathsf{T}$ contain the signed flow rates per unit span. The opening with the higher average static pressure is selected as the inlet. Its normal kinetic energy contributes to the driving head, while its tangential kinetic energy is assumed to dissipate on entry. With a uniform outlet jet ($\alpha_j = 1$), the channel energy balance is

$$\left|\Delta\overline{p}_k\right| + \frac{\rho}{2}\left\langle V_{n,\mathrm{in}}^2 \right\rangle = \Delta p_{f,k}(Q_k) + \frac{\rho}{2}\left(\frac{Q_k}{H_k}\right)^2, \quad (28)$$

where $\Delta\overline{p}_k = \overline{p}_{k,1} - \overline{p}_{k,2}$, $Q_k = |q'_k|$, and $\Delta p_{f,k}$ is the nonnegative frictional pressure loss. The averaging operator is defined in Section 2.3. Hydraulic inversion determines the flow magnitude, with its direction set by $\Delta\overline{p}_k$. Entrance turning and downstream jet mixing are not resolved explicitly.

For fixed geometry, the aerodynamic system is $\mathbf{A}\mathbf{x} = \mathbf{b}_\infty + \mathbf{B}\mathbf{q}$, where $\mathbf{x}$ contains the source and vortex strengths, and $\mathbf{B}$ transfers channel flows to the boundary conditions. A single LU factorisation of $\mathbf{A}$ gives

$$\mathbf{A}\mathbf{x}_0 = \mathbf{b}_\infty, \qquad \mathbf{A}\mathbf{X}_q = \mathbf{B}, \quad (29)$$

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{X}_q\mathbf{q}. \quad (30)$$

Here, $\mathbf{x}_0$ is the impermeable-airfoil solution, and each column of $\mathbf{X}_q$ represents the response to unit flow through one channel.

The surface velocities are therefore

$$\mathbf{V}_t = \mathbf{V}_{t,0} + \mathbf{T}_q\mathbf{q}, \quad (31)$$

$$\mathbf{V}_n = \mathbf{M}\mathbf{q}, \quad (32)$$

where $\mathbf{V}_{t,0} = \mathbf{v}_{\infty,t} + \mathbf{C}\mathbf{x}_0$, $\mathbf{T}_q = \mathbf{C}\mathbf{X}_q$, $\mathbf{C}$ maps aerodynamic strengths to induced tangential velocity, and $\mathbf{M}$ is the conservative opening-transfer operator. The Bernoulli relation, Eq. (10), then gives quadratic panel pressures. Linear opening averaging preserves this dependence, allowing the opening-pressure coefficients to be precomputed. This reduction is exact within the discretised model and avoids repeated panel solves without truncating the pressure relation.

The remaining nonlinear unknowns are the dimensionless channel flows $\mathbf{z}$, defined by

$$\mathbf{q} = \mathbf{S}\mathbf{z}, \qquad \mathbf{S} = V_\infty \,\mathrm{diag}(H_1, \dots, H_M). \quad (33)$$

For each trial $\mathbf{z}$, the reduced aerodynamic model provides the opening pressures and inlet normal kinetic heads. Hydraulic inversion gives $\mathbf{q}_{\mathrm{hyd}}(\mathbf{z})$, and the coupled solution satisfies

$$\mathbf{R}(\mathbf{z}) = \mathbf{z} - \mathbf{S}^{-1}\mathbf{q}_{\mathrm{hyd}}(\mathbf{z}) = \mathbf{0}. \quad (34)$$

All channel flows are solved simultaneously because they interact through the external pressure field.

A damped Newton method uses

$$\mathbf{J}_R = \mathbf{I}_M - \mathbf{S}^{-1}\frac{\partial \mathbf{q}_{\mathrm{hyd}}}{\partial \mathbf{z}}, \quad (35)$$

$$\mathbf{J}_R(\mathbf{z}^{(n)})\Delta\mathbf{z} = -\mathbf{R}(\mathbf{z}^{(n)}), \quad (36)$$

$$\mathbf{z}^{(n+1)} = \mathbf{z}^{(n)} + \eta_n\Delta\mathbf{z}, \quad (37)$$

where an Armijo line search selects η_n. The analytical Jacobian includes aerodynamic and hydraulic derivatives on the active inlet and friction branches; no smooth derivative is assumed across the abrupt Reynolds-number switch. The full nonlinear residual is re-evaluated after each update.

Initial estimates are obtained from the impermeable-airfoil pressure field or nearby converged solutions. A trust-region least-squares method provides a fallback, while parameter sweeps may additionally use adaptive angle continuation, Reynolds-number continuation, coupling homotopy and pseudo-arclength continuation. A solution is accepted only when the original nonlinear residual, channel energy balance and Reynolds-number regime conditions are satisfied at the requested operating point.

2.6. Verification and aerodynamic coefficients

A converged reduced solution is independently evaluated using a full panel solve and the original hydraulic inversion. Acceptance requires a sufficiently small dimensionless flow mismatch, a consistent channel energy balance and an admissible flow direction. Failed cases are recorded as nonconverged.

The pressure-force contribution of panel i is

$$\Delta C_{x,i} = -C_{p,i}n_{x,i}\frac{S_i}{c}. \quad (38)$$

$$\Delta C_{y,i} = -C_{p,i}n_{y,i}\frac{S_i}{c}. \quad (39)$$

The total force coefficients are

$$C_x = \sum_{i=1}^{N}\Delta C_{x,i}. \quad (40)$$

$$C_y = \sum_{i=1}^{N}\Delta C_{y,i}. \quad (41)$$

Lift and drag coefficients follow by rotation through the angle of attack α:

$$C_L = C_y\cos\alpha - C_x\sin\alpha. \quad (42)$$

$$C_D = C_x\cos\alpha + C_y\sin\alpha. \quad (43)$$

Using a clockwise-positive moment convention, the moment coefficient about $(x_{\mathrm{ref}}, y_{\mathrm{ref}})$ is

$$C_M = -\sum_{i=1}^{N} \left[\frac{x_i - x_{\mathrm{ref}}}{c} \Delta C_{y,i} - \frac{y_i - y_{\mathrm{ref}}}{c} \Delta C_{x,i} \right]. \tag{44}$$

These coefficients represent integrated pressure loads over the discretised panel boundary. They exclude viscous shear and an explicit additional pore momentum-flux reaction.

3. Results and discussion

The aerodynamic behaviour of the three porous-airfoil configurations was investigated over an angle-of-attack range of $-5°$ to $15°$. Three pore heights, $H = 8$, 9 and 10 mm, were considered for the horizontal, vertical and horizontal–vertical porous airfoils. The corresponding impermeable (solid) airfoil panel-method solution and the inviscid XFOIL prediction are included as reference cases.

The effect of the pores is first examined through the lift and moment coefficients over the angle-of-attack range. The observed aerodynamic trends are then interpreted using the corresponding pressure and velocity field differences. A detailed flow-field comparison is presented at $\alpha = 10°$ for a pore height of $H = 9$ mm.

3.1. Panel method validation against XFOIL

The impermeable source–vortex panel-method solution is compared with the corresponding inviscid XFOIL prediction in Figures. 2 and 3. Both methods show similar trends in lift and pitching moment over the investigated angle-of-attack range.

This agreement provides a useful reference for assessing the aerodynamic effects introduced by the pores. Small differences between the panel-method and XFOIL predictions are expected because the two approaches use different numerical formulations, although both represent inviscid external flow.

The impermeable panel-method solution is therefore used as the primary reference when evaluating the porous-airfoil configurations. This ensures that the porous and solid airfoils are compared using the same geometry, panel discretisation, aerodynamic formulation and operating conditions.

3.2. Effect of pores on lift

Figure 2 compares the variation of lift coefficient with angle of attack for the three porous-airfoil configurations and pore heights. The horizontal porous airfoil produces the largest change in lift. At positive angles of attack, its lift coefficient is higher than that of the impermeable reference, and the difference becomes more pronounced as the angle of attack increases. The effect also strengthens with pore height, with $H = 10$ mm producing the highest lift at larger positive angles of attack. At negative angles of attack, increasing the pore height similarly increases the magnitude of the negative lift.This nearly symmetric lift response about the zero-lift condition is consistent with the symmetric NACA 0018 geometry.

For the horizontal configuration, both ends of each pore are located on the same airfoil surface. The streamwise variation in surface pressure therefore creates a pressure difference between the two ends and drives internal flow from the locally higher-pressure region towards the lower-pressure region. This produces suction at one end and blowing at the other, changing the local surface-normal velocity and redistributing the external pressure and velocity fields along the airfoil surface. At positive angle of attack, the different pressure distributions over the suction and pressure surfaces lead to a net redistribution of aerodynamic loading that increases lift.

The stronger dependence on pore height is consistent with the reduction in hydraulic resistance as the pore becomes larger. For a given pressure difference, a larger pore height permits a greater internal flow and therefore a stronger interaction between the pore flow and the external aerodynamic field.

The vertical porous airfoil shows a distinctly different behaviour. At positive angles of attack, its lift coefficient is generally lower than that of the impermeable airfoil, while the three pore-height cases remain relatively close over most of the investigated range.

In this configuration, each pore connects the pressure surface to the suction surface. The resulting pressure difference drives internal flow between the two surfaces and partially reduces the pressure difference responsible for lift. This pressure equalisation is consistent with the lower lift coefficient obtained for the vertical porous airfoil. The small separation between the pore-height cases also indicates that this configuration is less sensitive to pore height than the horizontal arrangement.

The horizontal–vertical porous airfoil remains comparatively close to the impermeable reference. At positive angles of attack, its lift coefficient is generally slightly lower than that of the solid airfoil, and the three pore-height cases remain closely grouped.

This behaviour reflects the simultaneous influence of the horizontal and vertical pore arrangements. The horizontal pores redistribute flow between different chordwise locations on the same surface, whereas the vertical pores transfer flow between the pressure and suction surfaces. The lift-enhancing redistribution associated with the horizontal pores is therefore partly offset by the pressure equalisation introduced by the vertical pores.

The horizontal and vertical pores remain physically separate and hydraulically independent. However, their aerodynamic effects interact through the external pressure field, since changes produced by one set of pores alter the surface pressures that drive the other set. Consequently, the response of the horizontal–vertical porous airfoil is not a simple superposition of the two individual configurations.

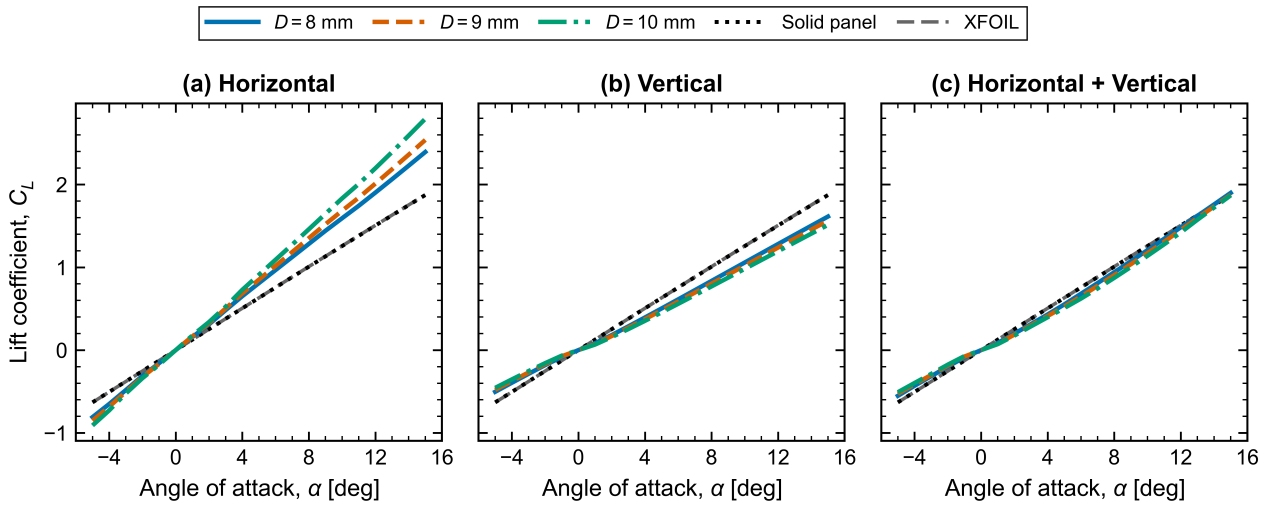


Figure 2: Lift coefficient as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Results are shown for pore heights $H = 8$, 9 and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

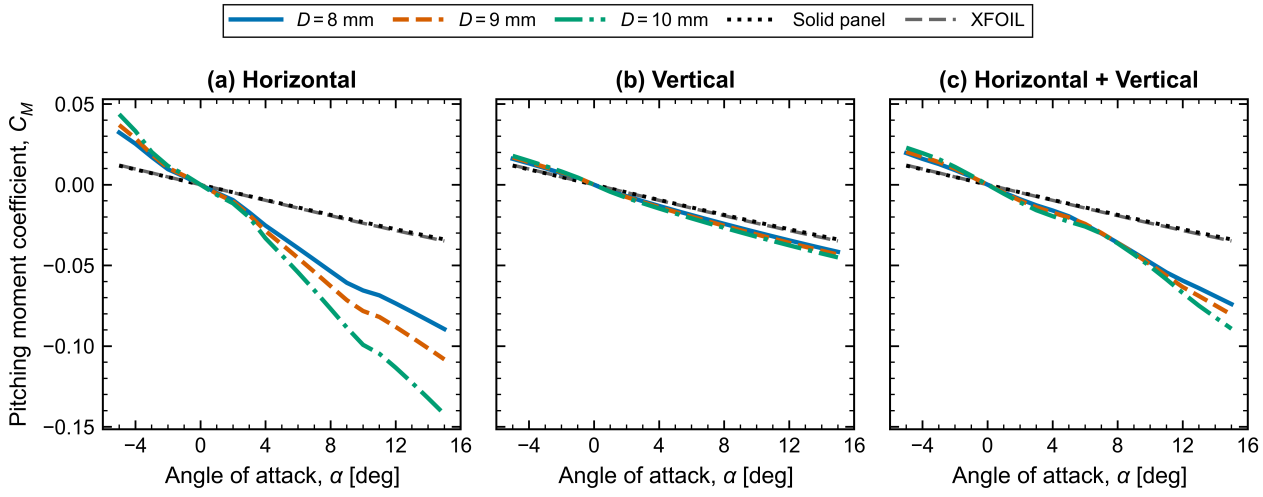


Figure 3: Pitching-moment coefficient about the quarter chord as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Results are shown for pore heights $H = 8$, 9 and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

Overall, the results show that the aerodynamic effect of the pores is governed mainly by their orientation, placement and the pressure difference between the connected surface locations. Pore height also influences the response, but its effect is strongest for the horizontal configuration and considerably weaker for the vertical and horizontal–vertical arrangements.

3.3. Effect of pores on pitching moment

The corresponding moment coefficients are presented in Figure. 3. The horizontal porous airfoil produces the largest increase from the impermeable-airfoil response. As the positive angle of attack increases, the pitching-moment coefficient becomes progressively more negative. The magnitude of this change also increases with pore height, with $H = 10$ mm producing the largest negative pitching moment at higher positive angles of attack.

This behaviour is associated with the chordwise redistribution of surface pressure caused by the horizontal pores. Since the two ends of each pore are located at different chordwise positions along a given airfoil surface, the pressure changes induced by the internal flow act at different distances from the quarter-chord reference point. These local pressure variations therefore contribute differently to the pitching moment. Increasing the pore height strengthens the internal flow and the resulting pressure redistribution, leading to a larger change in the pitching-moment coefficient.

The increase in lift obtained with the horizontal porous airfoil is therefore accompanied by a larger negative pitching

moment. This represents an important aerodynamic trade-off, since the configuration that provides the greatest lift enhancement also produces the largest change in longitudinal loading.

The vertical porous airfoil produces considerably smaller changes in moment, and the three pore-height cases remain closely grouped over most of the investigated angle-of-attack range. This behaviour is consistent with the weaker dependence on pore height observed in the lift results.

For the vertical configuration, the two ends of each pore are positioned on the upper and lower surfaces at approximately the same chordwise location. The resulting pressure redistribution therefore occurs mainly between the pressure and suction surfaces rather than between widely separated positions along the chord. The corresponding change in the chordwise pressure distribution is smaller than for the horizontal pores, resulting in a weaker pitching-moment response.

The horizontal–vertical porous airfoil exhibits behaviour between that of the individual horizontal and vertical configurations. The pitching moment becomes increasingly negative with positive angle of attack, although the departure from the impermeable-airfoil solution remains smaller than for the horizontal porous airfoil. The influence of pore height also becomes more noticeable at higher positive angles of attack.

In this configuration, the horizontal pores redistribute loading between different chordwise positions, while the vertical pores mainly redistribute pressure between the upper and lower surfaces. The two pore arrangements remain physically separate and hydraulically independent, but their aerodynamic effects interact through the external pressure field. The resulting pitching-moment variation therefore reflects the combined influence of these two pressure-redistribution mechanisms.

Overall, the pitching-moment results show that the effect of the pores depends not only on the extent of the pressure redistribution, but also on where it occurs along the chord. The horizontal porous airfoil produces the largest change in moment because the associated pressure variations occur at widely separated chordwise locations, creating a stronger moment about the quarter-chord reference point. In contrast, the vertical porous airfoil produces a smaller change because its pressure redistribution occurs mainly between the upper and lower surfaces at similar chordwise positions.

3.4. Flow field characteristics of porous airfoils

To examine how the pores alter the external flow, the pressure and velocity fields of the porous airfoils are compared with those of the corresponding impermeable airfoil at $\alpha = 10°$ and $H = 9$ mm.

The comparison is presented using pressure- and velocity-difference contours rather than the absolute flow fields. The absolute contours are dominated by the overall aerodynamic behaviour of the airfoil and therefore make the local influence of the pores more difficult to identify. Subtracting the impermeable-airfoil solution isolates the changes introduced by the pore flow, allowing the location, direction and relative strength of the resulting pressure and velocity variations to be examined more clearly.

3.4.1. Pressure field characteristics

Figure 4 presents the pressure difference between the porous and impermeable airfoils,

$$\Delta p = p_{\text{porous}} - p_{\text{solid}}, \tag{45}$$

for the three pore arrangements.

The pressure difference is shown instead of the absolute pressure field to isolate the changes introduced by the pores. In the absolute solution, the pressure distribution is dominated by the overall airfoil loading and the strong pressure gradients near the leading edge. Subtracting the impermeable-airfoil solution makes the local changes associated with the pore flow easier to identify.

The strongest pressure differences occur near the pore ends, where the internal flow directly influences the surrounding external flow. In several locations, neighbouring positive and negative values of Δp form a dipole-like pattern. This is associated with suction at one end of a pore and blowing at the other, producing local regions of reduced and increased pressure relative to the impermeable airfoil.

The effect of these pressure changes on lift depends on where they occur on the airfoil. At positive angle of attack, a reduction in pressure on the upper suction surface strengthens the suction and contributes to higher lift, whereas an increase in pressure has the opposite effect. On the lower pressure surface, an increase in pressure contributes positively to lift, while a pressure reduction tends to decrease it.

For the horizontal porous airfoil, the pressure differences are mainly concentrated around the chordwise pore ends. Because both ends of each pore lie on the same airfoil surface, the internal flow redistributes pressure between different chordwise locations. The combined pressure changes on the upper and lower surfaces increase the pressure difference across the airfoil, consistent with the higher lift coefficient observed for this configuration.

The vertical porous airfoil produces a different pattern because each pore connects the pressure surface to the suction surface. Flow is therefore driven from the higher-pressure region towards the lower-pressure region, producing local pressure changes at both pore ends. At the same time, this cross-surface flow partially reduces the pressure difference between the two surfaces, which is consistent with the lower lift coefficient obtained for the vertical configuration.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores redistribute pressure along the airfoil surfaces, while the vertical pores transfer flow between the pressure and suction surfaces. Although the two pore systems remain physically separate and hydraulically independent, their aerodynamic effects interact through the external pressure field. Consequently, the pressure equalisation associated with the

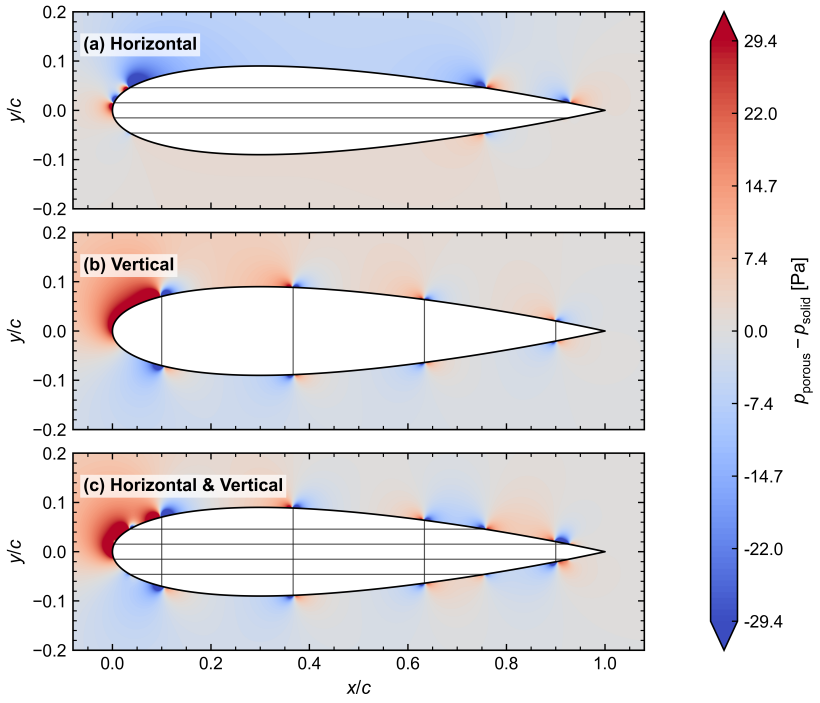


Figure 4: Pressure-field difference between the porous and impermeable airfoils at $\alpha = 10°$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in pressure relative to the impermeable airfoil, while negative values indicate a reduction.

vertical pores partly counteracts the pressure redistribution produced by the horizontal pores, giving an overall aerodynamic behaviour closer to that of the impermeable airfoil.

3.4.2. Velocity field characteristics

The velocity field is presented as the percentage difference relative to the impermeable-airfoil solution,

$$\Delta|\boldsymbol{V}|_{\%} = \frac{|\boldsymbol{V}|_{\text{porous}} - |\boldsymbol{V}|_{\text{solid}}}{|\boldsymbol{V}|_{\text{solid}}} \times 100. \tag{46}$$

This comparison isolates the changes introduced by the pores from the dominant background flow around the airfoil. The largest velocity differences occur close to the pore ends, where the imposed surface-normal flow locally perturbs the external velocity field. Adjacent regions of acceleration and deceleration are visible around several pores and correspond closely to the pressure patterns shown in Figure. 4.

The velocity and pressure fields show the expected inverse relationship described by Bernoulli's equation. Regions where the flow accelerates are generally associated with a reduction in static pressure, whereas regions of lower velocity correspond to higher static pressure. The velocity contours therefore help explain how the pore-induced changes in the local flow field lead to the observed pressure redistribution and, consequently, changes in the aerodynamic forces and pitching moment.

For the horizontal porous airfoil, alternating regions of acceleration and deceleration are concentrated around the chordwise pore ends. Since both ends of each pore lie on the same airfoil surface, the internal flow redistributes the external velocity between different chordwise locations. The combined changes on the suction and pressure surfaces strengthen the pressure difference across the airfoil, consistent with the increase in lift and the larger change in pitching moment observed for this configuration.

The vertical porous airfoil shows a different velocity pattern. Flow between the pressure and suction surfaces changes the external velocity on both sides of the airfoil and reduces part of the velocity contrast between them. This behaviour is consistent with the partial pressure equalisation discussed previously and the corresponding reduction in lift. A broader region of velocity difference is also visible near the leading edge, where the external flow experiences strong pressure and velocity gradients.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores produce local chordwise acceleration and deceleration, while the vertical pores redistribute flow between the pressure and suction surfaces. Although the two pore arrangements remain physically separate and hydraulically independent, their aerodynamic effects interact through the external flow field. The resulting velocity distribution therefore reflects the combined influence of both mechanisms rather than a simple superposition of the individual configurations.

Overall, the velocity contours support the trends observed in the pressure field and aerodynamic coefficients. The horizontal pores primarily redistribute the flow along the airfoil surfaces, whereas the vertical pores transfer flow

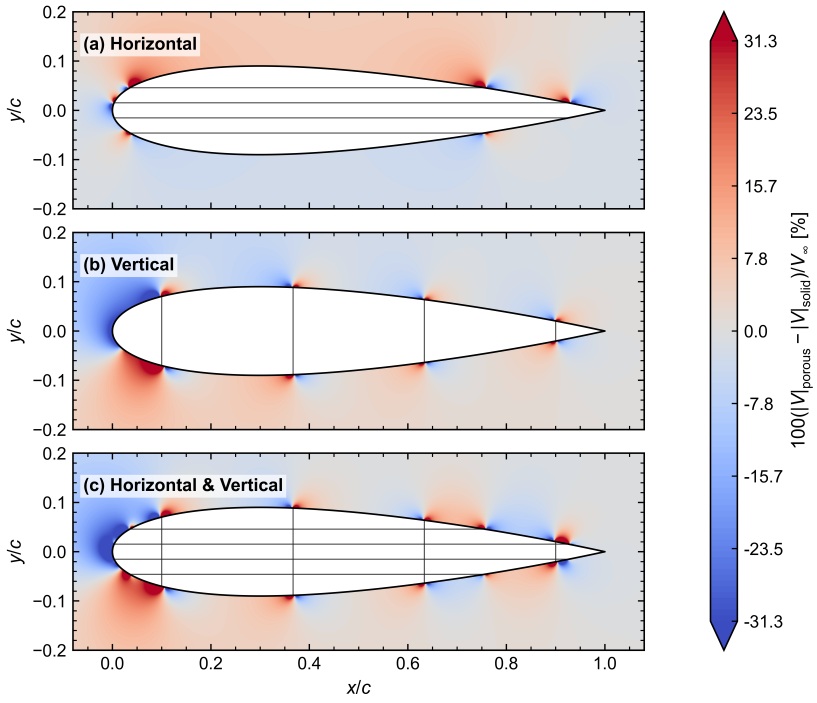


Figure 5: Percentage difference in velocity magnitude between the porous and impermeable airfoils at $\alpha = 10°$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in velocity magnitude, while negative values indicate a reduction relative to the impermeable airfoil.

between the pressure and suction surfaces. These different flow mechanisms contribute to the contrasting lift and pitching-moment behaviour of the three porous-airfoil configurations.

4. Conclusion and future work

A reduced-order aero–hydraulic framework has been developed to investigate NACA 0018 airfoils containing pressure-driven internal channels. The external flow is represented using a source–vortex panel method and is coupled to finite-length rectangular-channel models through the pore-normal velocity. The main conclusions of the present study are:

- The aerodynamic response of a porous airfoil depends strongly on the orientation, location and connectivity of the internal channels. Porosity alone is therefore insufficient to describe the resulting aerodynamic behaviour.
- Horizontal chordwise channels produce the strongest aerodynamic modification among the configurations considered. By connecting different chordwise locations on the same airfoil surface, these channels redistribute pressure without directly equalising the pressure and suction surfaces.
- At positive angles of attack, the horizontal configuration increases lift relative to the impermeable airfoil. The magnitude of this effect increases with channel height, demonstrating the sensitivity of the aerodynamic response to the hydraulic resistance of the internal passages.
- The lift enhancement produced by the horizontal channels is accompanied by a more negative pitching moment. This demonstrates that improvements in lift cannot be considered independently of changes in longitudinal aerodynamic loading.
- Vertical channels exhibit a fundamentally different behaviour. Connecting the pressure and suction surfaces allows pressure-driven flow between the two sides of the airfoil, producing partial pressure equalisation and consequently reducing lift relative to the impermeable configuration at positive angles of attack.
- The combined horizontal–vertical configuration does not behave as a simple superposition of the two individual arrangements. Although the channel systems are hydraulically independent, they interact through the external pressure field. The pressure equalisation associated with the vertical channels therefore partly offsets the lift-enhancing effect of the horizontal channels.
- Pressure- and velocity-difference fields show that the strongest aerodynamic modifications occur locally around the channel openings. Suction and blowing generated by the internal flow alter the surrounding

external velocity and pressure fields, which subsequently modify the integrated aerodynamic forces and pitching moment.

- The results demonstrate that the pressure difference between connected surface locations is a key driver of porous-airfoil behaviour. Channel placement, orientation and height should therefore be considered simultaneously when designing internal porous networks.

- The developed framework provides a computationally inexpensive method for screening multiple internal-channel configurations before progressing to substantially more expensive viscous CFD simulations or experimental investigations.

The present results should be interpreted as inviscid aerodynamic trends. The external panel method does not resolve boundary layers, transition, skin-friction drag, flow separation or stall, and the pressure-based streamwise-force coefficient should therefore not be interpreted as the total drag of the porous airfoil. These limitations define the next stage of the investigation.

4.1. Future work

Future development will focus on:

- incorporating a viscous boundary-layer model into the panel-method framework to estimate skin-friction drag, transition and the influence of viscous–inviscid interaction;

- extending the design space beyond the three prescribed configurations by optimising channel location, orientation, height and connectivity to identify improved internal porous-network topologies;

- investigating selected configurations using higher-fidelity unsteady CFD to resolve pore-mouth flow, boundary-layer interaction, separation and wake dynamics that cannot be represented by the present reduced-order formulation;

- validating the predicted pressure redistribution and aerodynamic trends experimentally using manufactured porous-airfoil models; and

- extending the analysis towards three-dimensional porous wings and application-relevant operating conditions once the underlying aero–hydraulic mechanisms have been established.

The present framework therefore provides the first stage of a hierarchical design process in which rapid reduced-order screening is used to identify promising internal-channel arrangements for subsequent optimisation, high-fidelity viscous simulation and experimental validation.

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